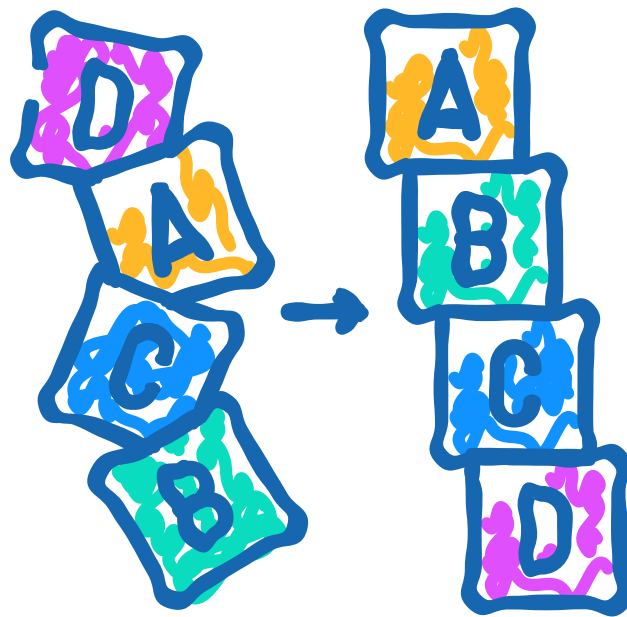
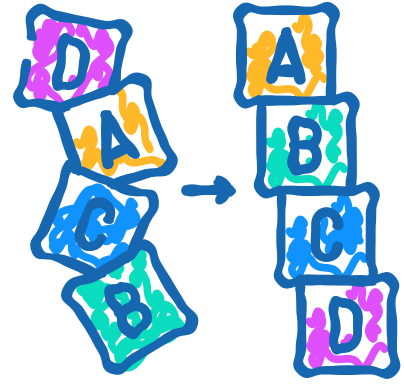


Data Structures and Algorithms





What is a data structure?

a way of organizing, storing, and managing data in a computer so it can be accessed and modified efficiently.

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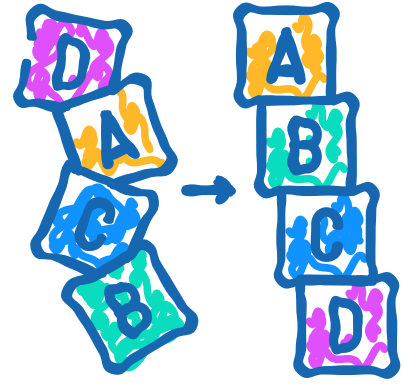
a systematic way to store and organize data for efficient use.

Why are data structures important?

Different tasks need different ways of storing data. A good data structure can make programs faster, use less memory, and make code easier to manage.



Stacks



What is a Stack?

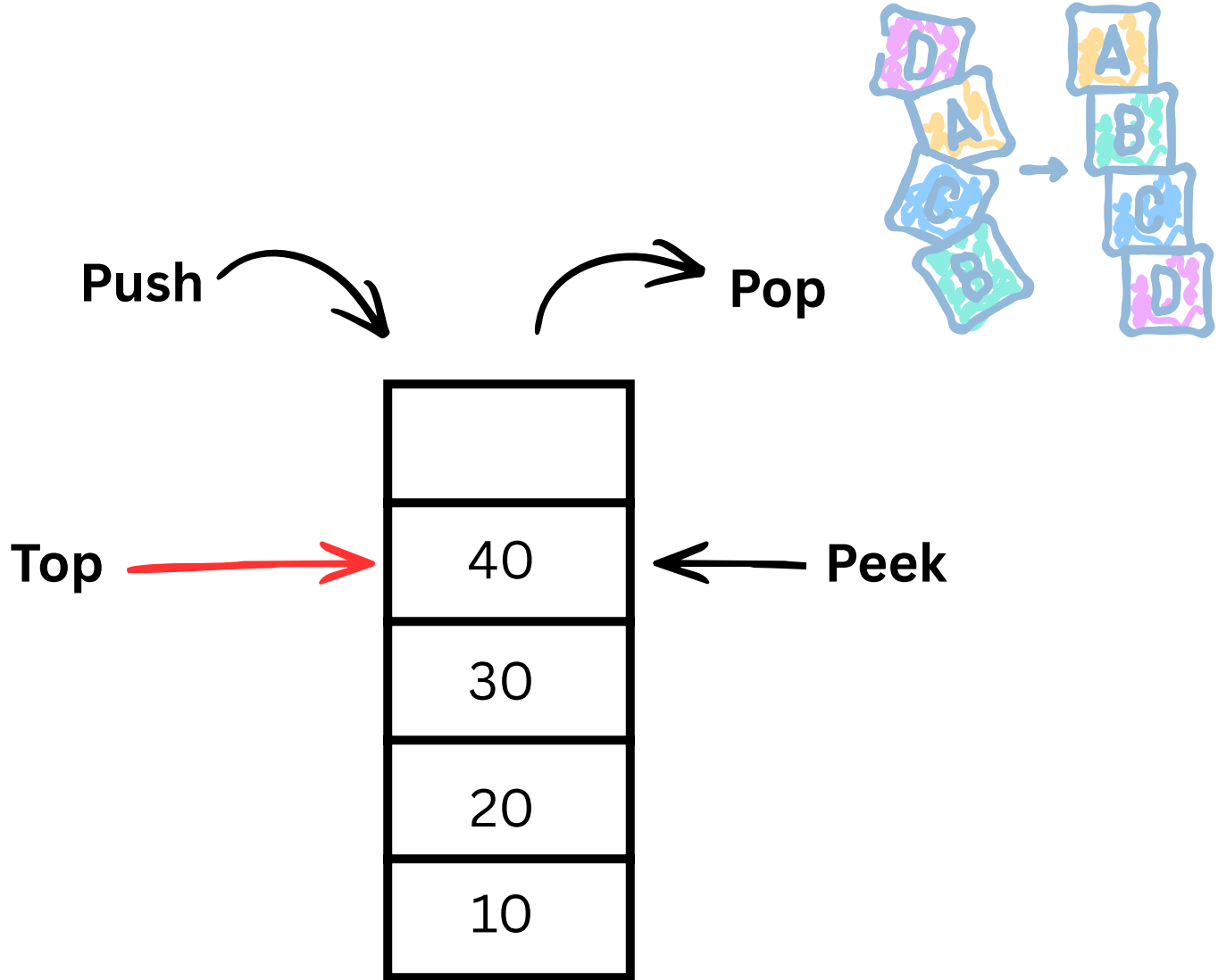
- A Stack is a linear data structure.
- It follows the LIFO (Last In, First Out) principle.
- The last element inserted is the first element removed.

Examples

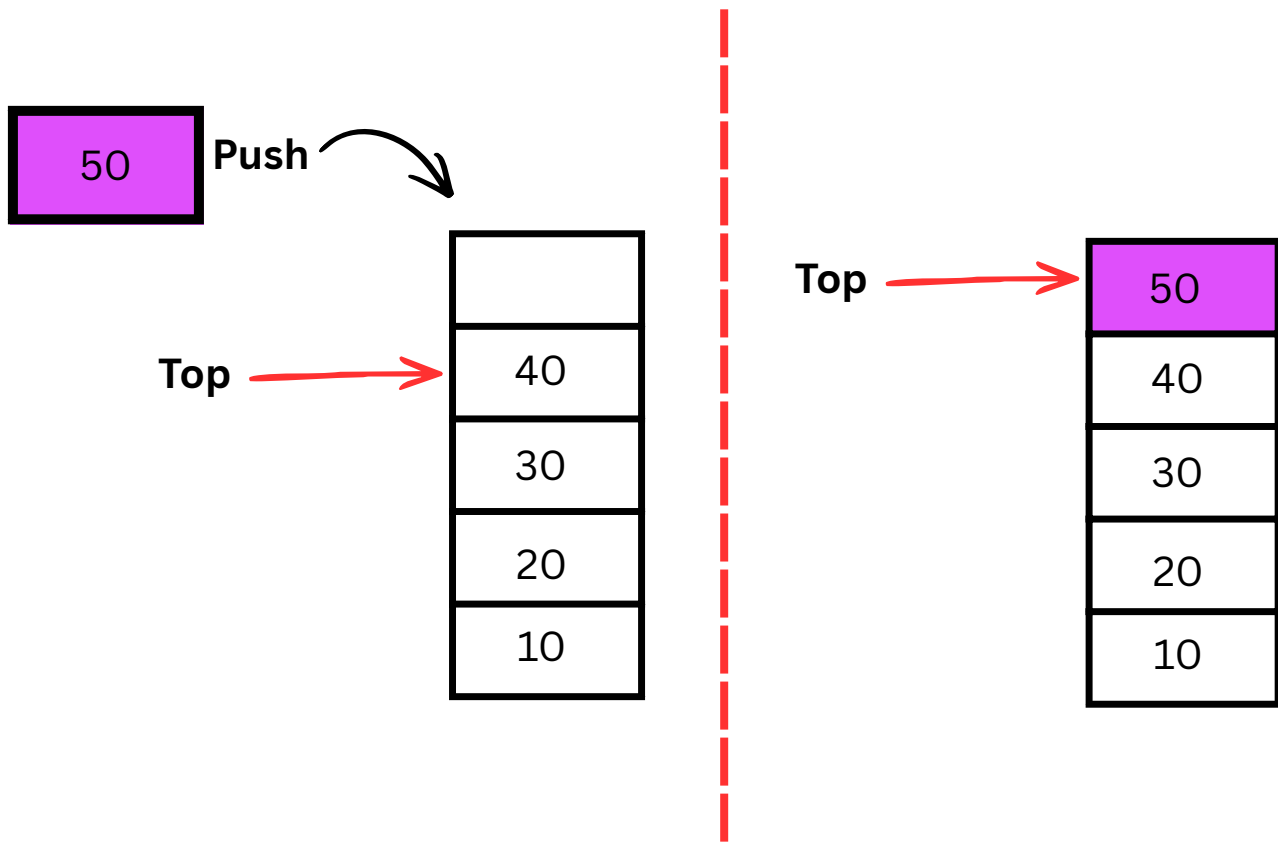
- Browser back button history.
- Undo operation in text editors.
- Function call stack in programming.

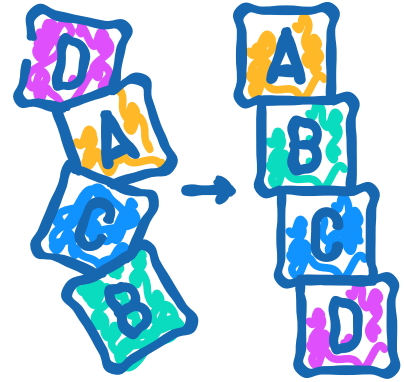
Basic Operations

- Push – Add an element to the top of the stack.
- Pop – Remove the top element from the stack.
- Peek/Top – View the top element without removing it.
- isEmpty – Check whether the stack is empty.
- isFull – Check whether the stack is full (for fixed-size implementations).

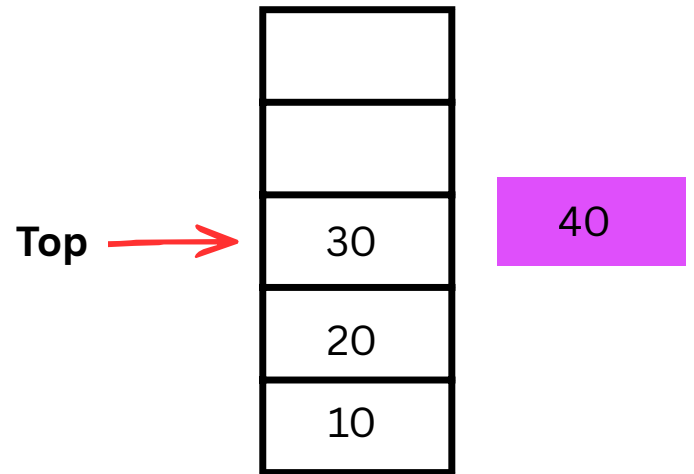
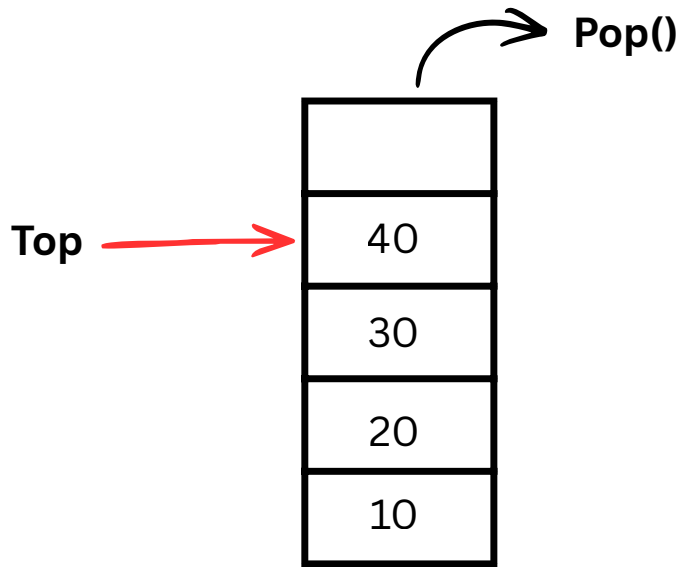


Add an element

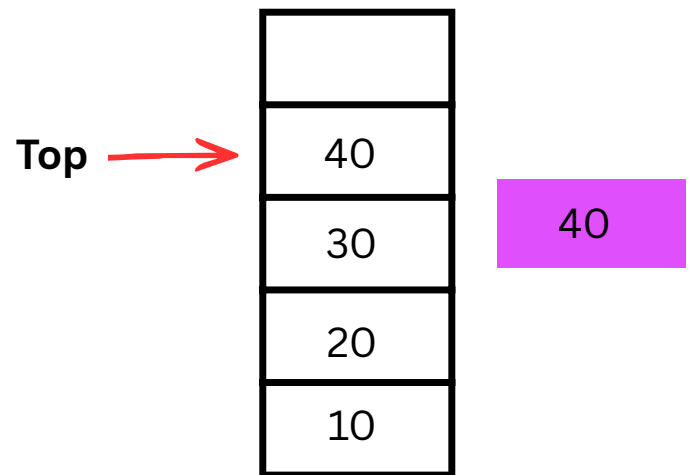
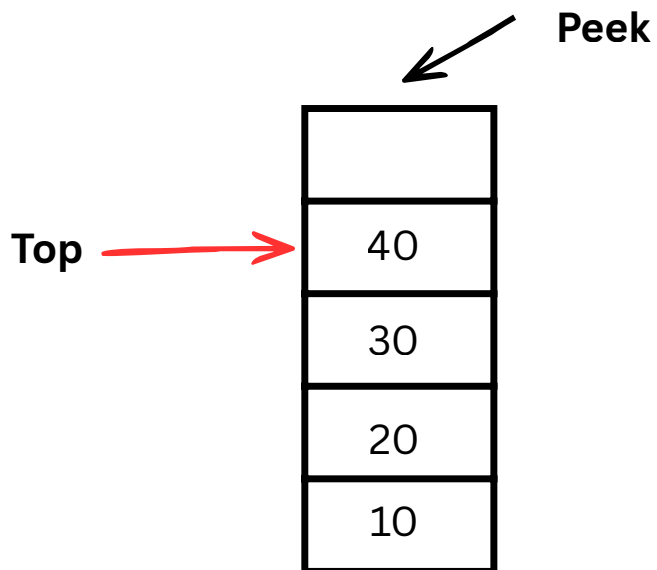


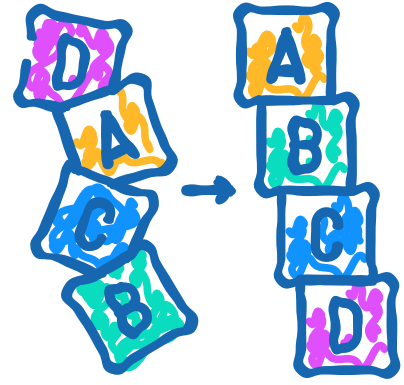


Remove the top element



View top element





Characteristics

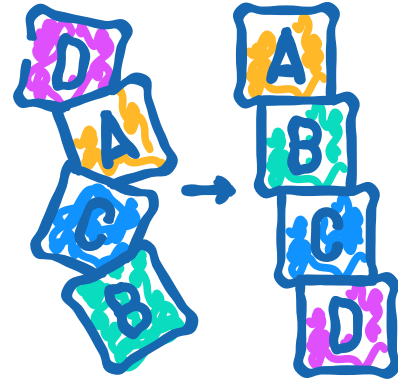
- Insertion and deletion occur only at one end called the Top.
- Simple and efficient data structure.
- Easy to implement using arrays or linked lists.

Advantages

- Simple implementation.
- Fast insertion and deletion ($O(1)$).
- Efficient memory usage in many applications.
- Supports recursive function execution.

Disadvantages

- Limited access to elements.
- Only the top element can be accessed directly.
- Stack overflow may occur if capacity is exceeded.

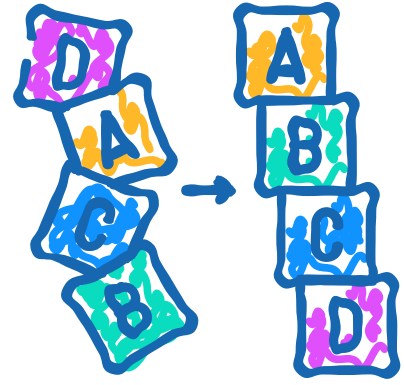


Applications of Stack

- Function call management (Call Stack).
- Recursive function execution.
- Expression evaluation and conversion.
- Parentheses matching.
- Undo/Redo operations.
- Browser history navigation.
- Backtracking algorithms.
- Depth First Search (DFS).
- String reversal.

Time Complexity

Operation	Time Complexity
Push	$O(1)$
Pop	$O(1)$
Peek/Top	$O(1)$
Search	$O(n)$



Types of Stacks

- Static Stack (Array-based)
- Dynamic Stack (Linked List-based)

Stack Overflow

- Occurs when attempting to push an element into a full stack.

Stack Underflow

- Occurs when attempting to pop an element from an empty stack.

Important

A Stack is a linear data structure that stores and retrieves data according to the Last In, First Out (LIFO) principle, making it useful for recursion, backtracking, expression processing, and memory management.